

Flag 1 – Install Flutter SDK

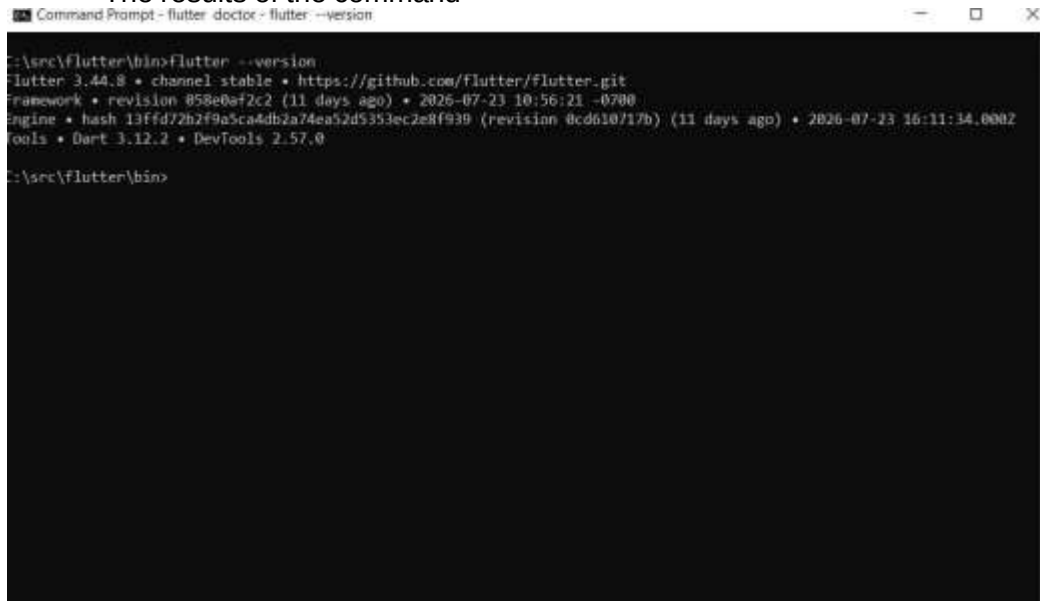
Objective: Install the Flutter SDK.

Verify:

`flutter --version`

Screenshot:

- The results of the command



```
Command Prompt - flutter doctor - flutter --version

C:\src\flutter\bin>flutter --version
Flutter 3.44.8 • channel stable • https://github.com/flutter/flutter.git
Framework • revision 058e0af2c2 (11 days ago) • 2026-07-23 10:56:21 -0700
Engine • hash 13ffd72b2f9a5ca4db2a74ea52d5353ec2e8f939 (revision 0cd61017b) (11 days ago) • 2026-07-23 16:11:34.000Z
Tools • Dart 3.12.2 • DevTools 2.57.0

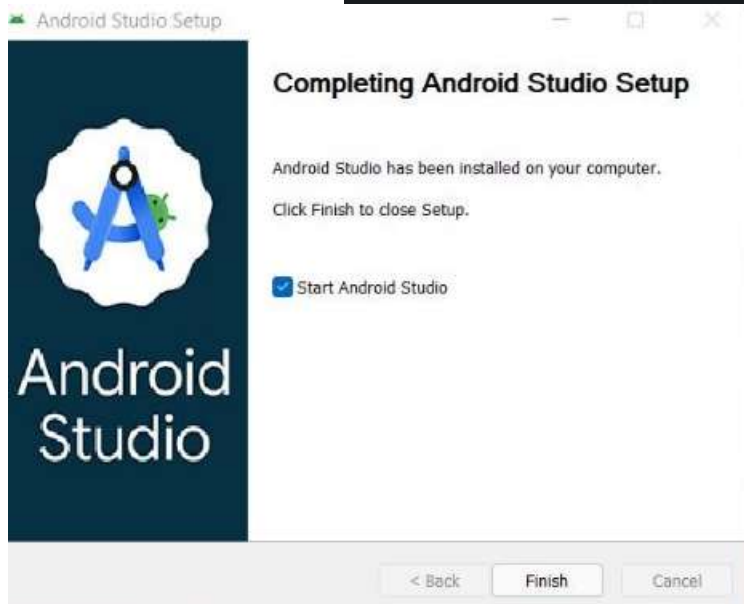
C:\src\flutter\bin>
```

Flag 2 – Install Android Studio

Objective: Install Android Studio.

Screenshot:

- Android Studio installed
- SDK Manager opened (or Welcome screen)

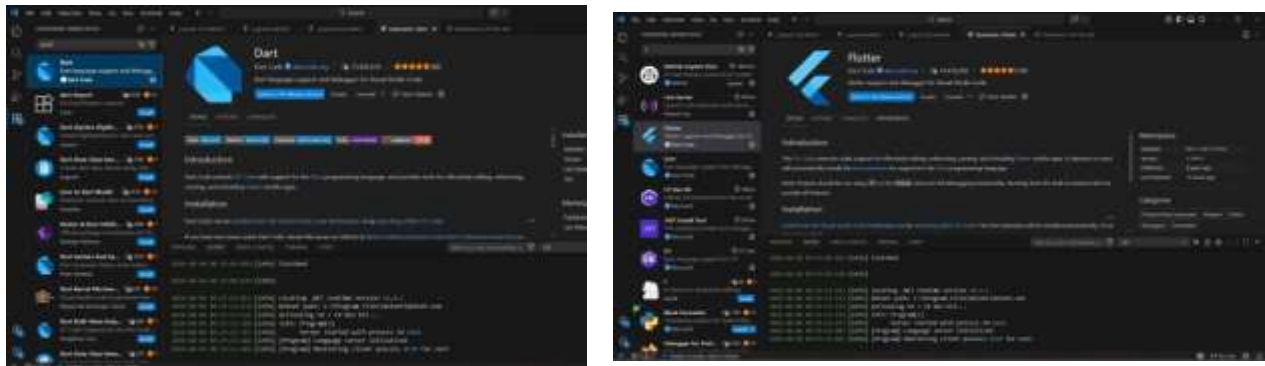


Flag 3 – Install VS Code & Flutter Extension

Objective: Install VS Code and the Flutter (which also installs Dart) extension.

Screenshot:

- Flutter extension installed
- Dart extension installed (or shown as installed)



Flag 4 – Run Flutter Doctor

Objective: Check whether everything is configured correctly.

Run:

flutter doctor

Screenshot:

- flutter doctor results

```
C:\Users\Ninja Turtles>flutter doctor
Flutter' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Ninja Turtles>C:\src\flutter\bin\flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[!] Flutter (Channel stable, 1.44.0, on Microsoft Windows [Version 10.0.21996.1], locale en-US)
    ! The Flutter binary is not on your path. Consider adding C:\src\flutter\bin to your path.
    ! The Dart binary is not on your path. Consider adding C:\src\flutter\bin to your path.
[x] Windows Version (11 Home Single Language 64-bit, Dev, 20H2)
[x] Android toolchain - develop for Android devices
    ! cmdline-tools component is missing
      Try installing or updating Android Studio.
      Alternatively, download the tools from https://developer.android.com/studio/command-line and make sure
      to set the ANDROID_HOME environment variable.
      See https://developer.android.com/studio/command-line for more details.
[x] Chrome - develop for the web
[x] Visual Studio - develop Windows apps (Visual Studio Community 2025 18.4.1)
    ! Visual Studio is missing necessary components. Please re-run the Visual Studio installer for the "Desktop
      development with C++" workload, and include these components:
        MSVC v142 - VS 2019 C++ x64/x86 build tools
          - If there are multiple build tool versions available, install the latest
        C++ Make tools for Windows
        Windows 10 SDK
[x] Connected device (3 available)
[x] Network resources

Doctor found issues in 3 categories.

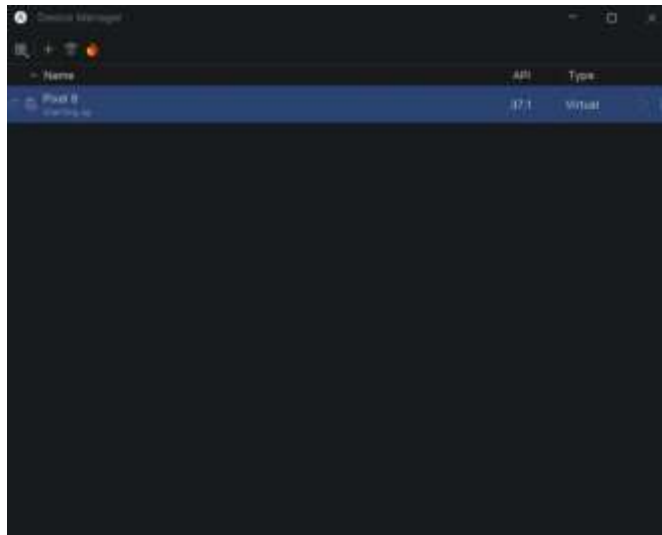
C:\Users\Ninja Turtles>
```

Flag 5 – Build Your Virtual Device

Objective: Create your first Android Emulator (AVD).

Screenshot:

- Device Manager
- Your newly created virtual device
- Emulator successfully booted



Flag 6 – Forge Your Project

Objective: Create your first Flutter project.

Run:

```
flutter create my_first_flutter_app
```

Screenshot:

- Successful creation of project

```
C:\Windows\system32\cmd.exe - flutter create C:\Users\Ninja Turtle\Documents\my_first_flutter_app - flutter create "C:\Users\Ninja Turtle\Documents\my_first_flutter_app"

C:\Users\Ninja Turtle>cd C:\src\Flutter\bin

C:\src\Flutter\bin>flutter create C:\Users\Ninja Turtle\Documents\my_first_flutter_app
Multiple output directories specified.

C:\src\Flutter\bin>flutter create "C:\Users\Ninja Turtle\Documents\my_first_flutter_app"
Creating project ..\..\..\Users\Ninja Turtle\Documents\my_first_flutter_app...
Resolving dependencies in "..\..\..\Users\Ninja Turtle\Documents\my_first_flutter_app"... (1.4s)
Downloading packages...
Got dependencies in "..\..\..\Users\Ninja Turtle\Documents\my_first_flutter_app".
Wrote 131 files.

All done!
You can find general documentation for Flutter at: https://docs.flutter.dev/
Detailed API documentation is available at: https://api.flutter.dev/
If you prefer video documentation, consider: https://www.youtube.com/c/flutterdev

In order to run your application, type:

$ cd ..\..\..\Users\Ninja Turtle\Documents\my_first_flutter_app
$ flutter run

Your application code is in ..\..\..\Users\Ninja Turtle\Documents\my_first_flutter_app\lib\main.dart.

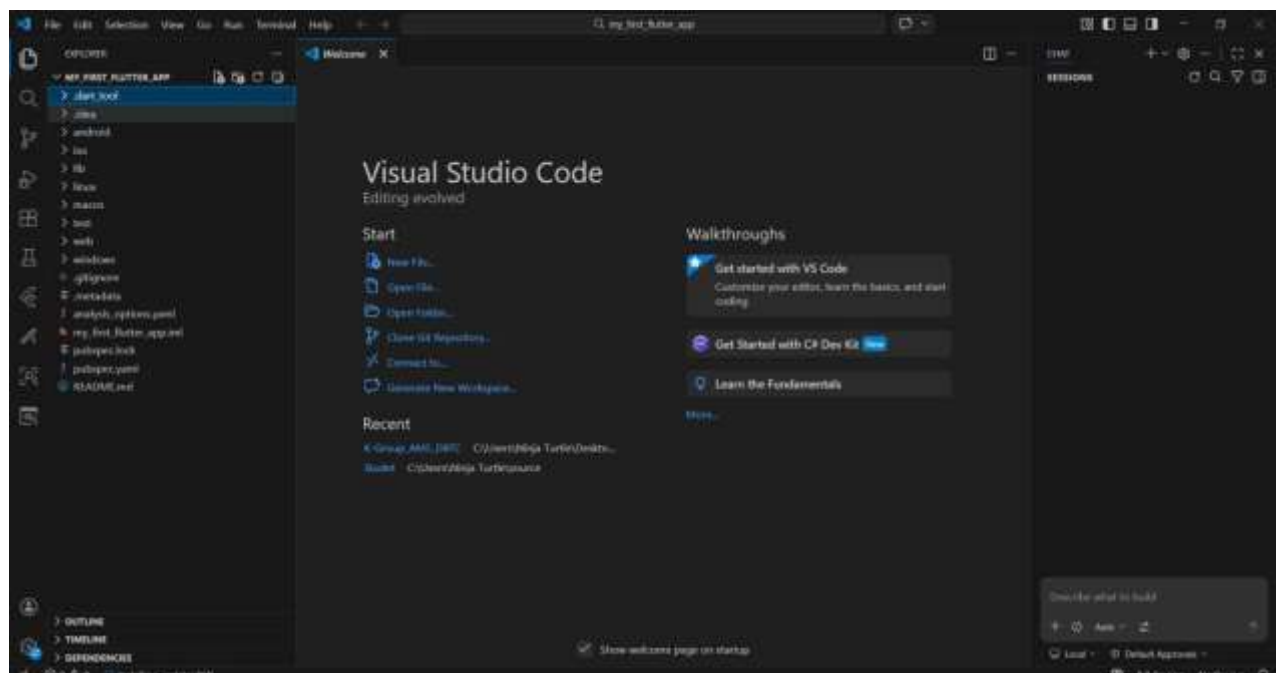
C:\src\Flutter\bin>
```

Flag 7 – Unlock the Workspace

Objective: Open the project in VS Code.

Screenshot:

- Your project opened in VS Code



Flag 8 – Launch

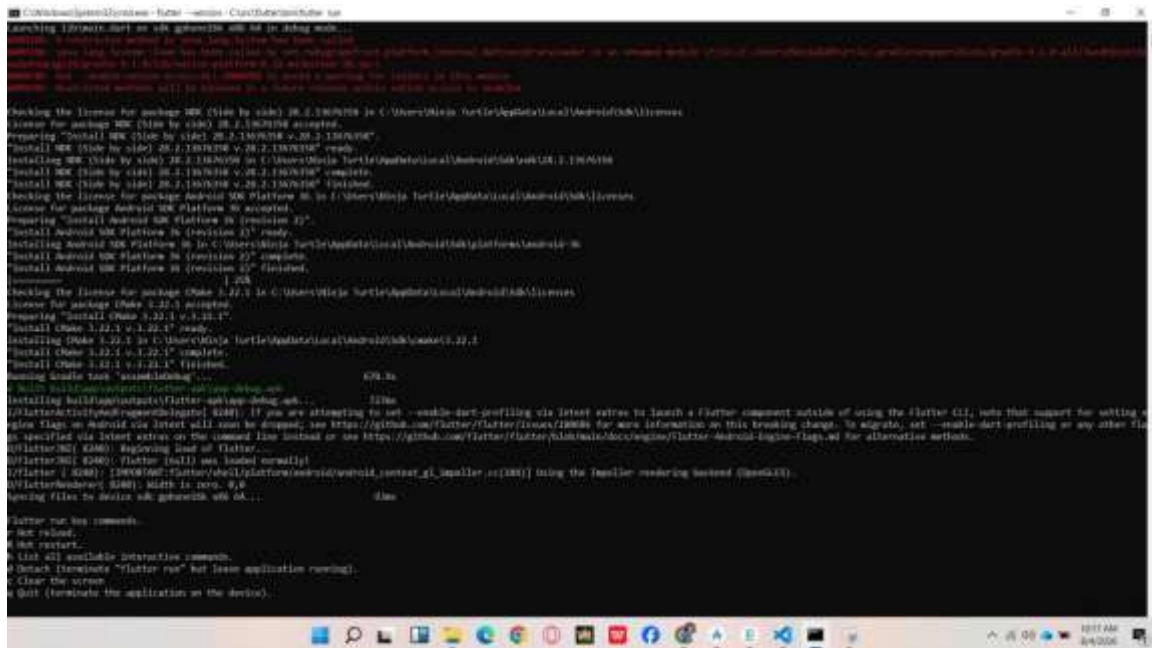
Objective: Run your Flutter application.

Run:

flutter run

Screenshot:

- Terminal
- Emulator running
- Default Flutter app

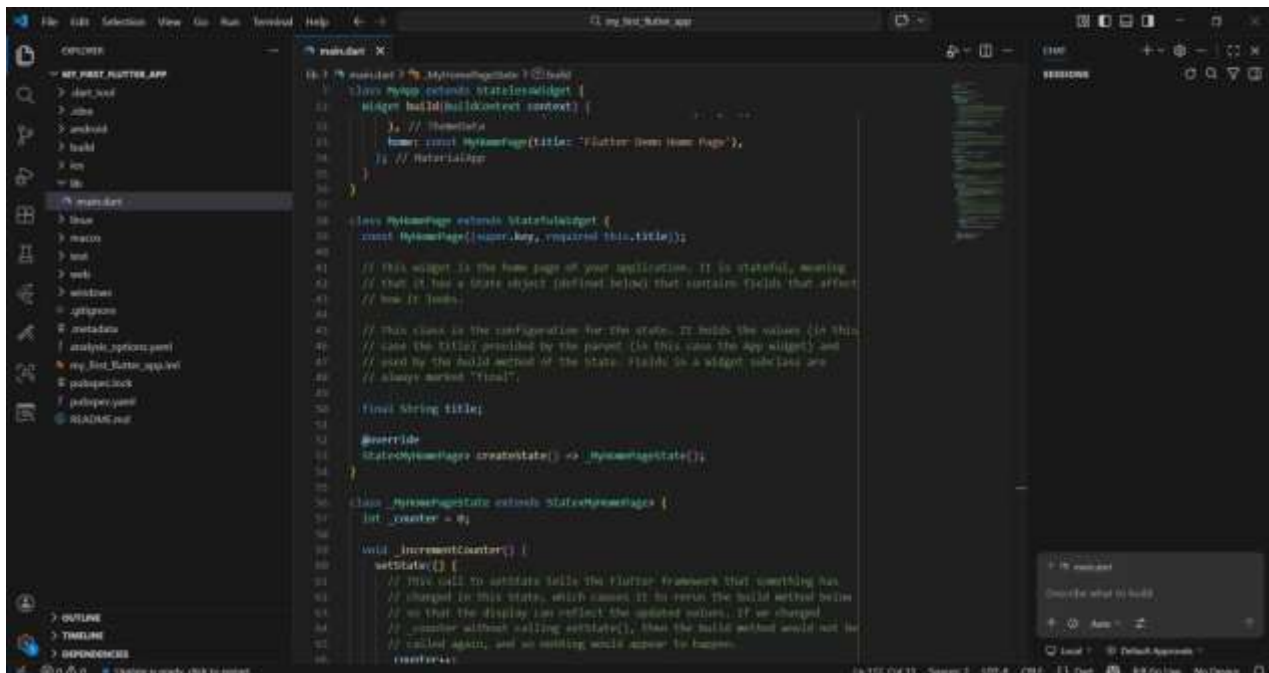


Flag 9 – Find the Entry Point

Figure out where the default Flutter page comes from.

Screenshot:

- The location of the file where the default page is from (the one showing on the emulator)



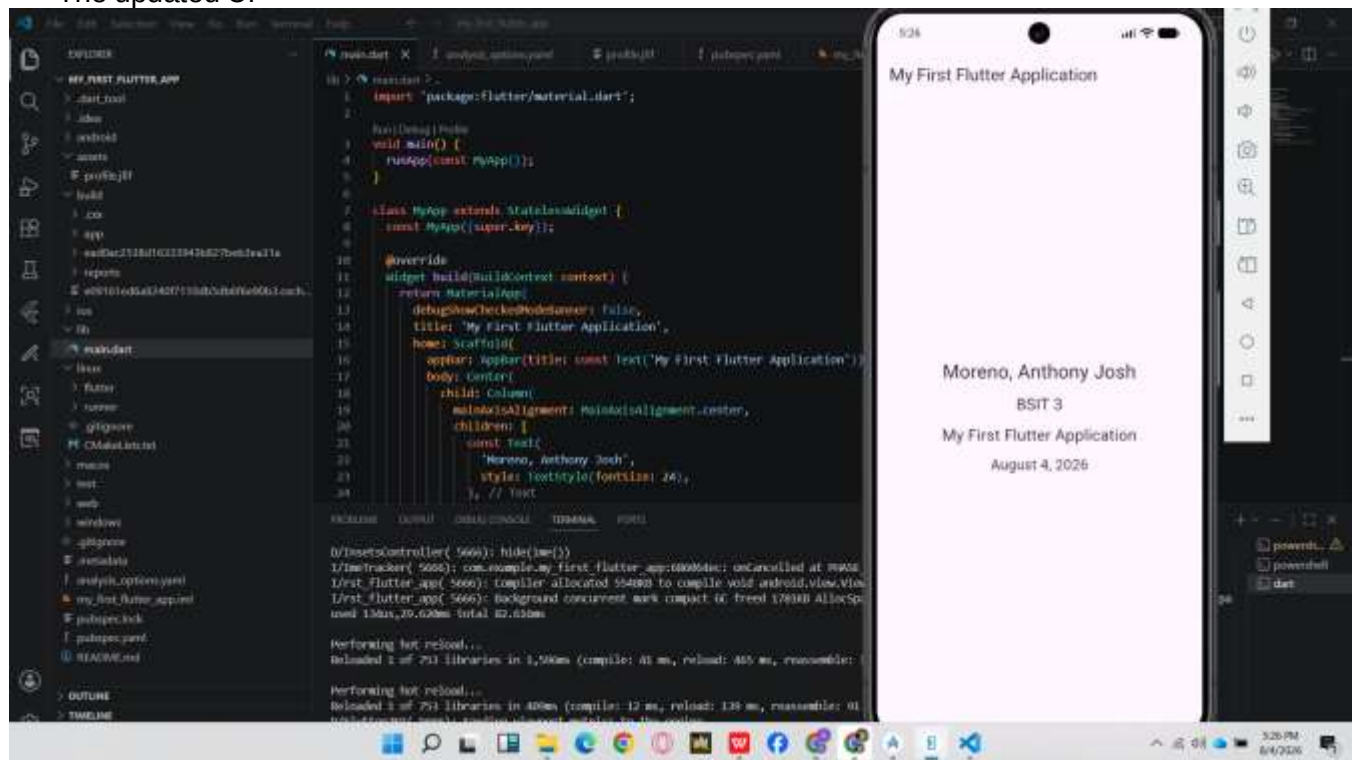
Flag 10 – Claim Your App

Replace the default screen with:

- Full Name
- Course & Section
- "My First Flutter Application"
- Today's Date

Screenshot:

- Your updated content of the code
- The updated UI



Flag 11 – Personalize It

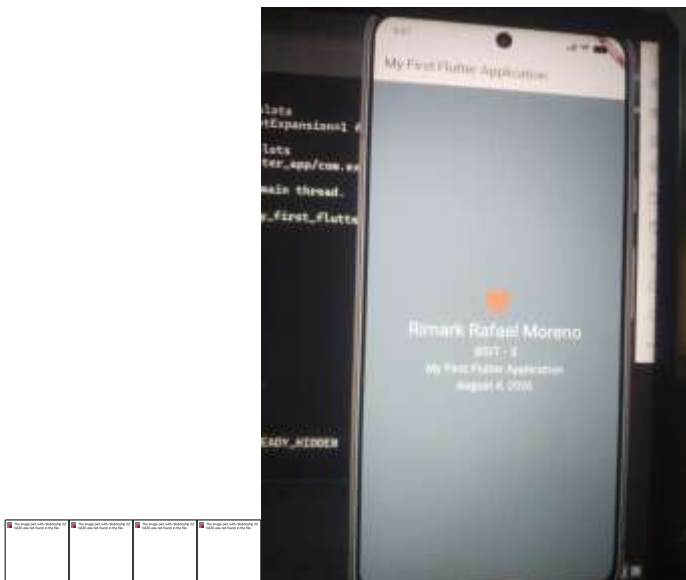
Research and implement at least **three** visual changes.

Examples:

- Change the background colour.
- Change the text colour.
- Centre the content.
- Increase font size.
- Add a border or rounded card.
- Change the font.
- Add an emoji or icon.

Screenshot:

- The code changed for personalizing
- The updated UI



Flag 12 – Add an Image

Research how to:

- Create an `assets` folder
- Register assets
- Display an image

Screenshot:

- The steps it took to add an image
- The updated UI

